Comparative Study of Abaca and Pineapple Leaf Fiber-Reinforced Epoxy Composites for Lightweight Drone Frame Plate Applications

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Chapter 1

INTRODUCTION

In the world of engineering and materials science, new materials are developed every year to make products and structures stronger, more durable, and lighter (mechanicalengineerings.com, 2025). In recent years, there has been a growing awareness of environmental concerns and a push towards more sustainable materials in various industries (addcomposites2024). This trend has led to increased interest among researchers in the study of natural fiber composites (NFCs), which offer a promising alternative to traditional synthetic fiber-reinforced materials. Composite materials have been used for thousands of years. At that time, they manufactured bricks using mud, which was made based on thousand-year-old technology. It was first used in 1500 B.C. Early Egyptians and Mesopotamian builders and artisans made strong and durable buildings using a mixture of mud and straw (Autodesk Inc. 2022). Today, the use of composite materials is rapidly increasing in almost every industry, including aerospace, automotive, civil engineering, and sports. Composite materials are engineered materials made by combining two or more constituent materials with different physical and chemical properties to create a new material with superior characteristics (ArchDaily.com). Composite materials can be made using various types of fibers: carbon fiber, glass fiber, and natural fibers. Natural fibers can be used as a sustainable alternative to synthetic fibers; they are renewable, biodegradable, Low Density, Low Cost, easier on processing equipment, and less harmful to the environment compared to glass or carbon fibers.

Natural fibers can be broadly classified into two categories: plant-based fibers and animal-based fibers (Textile Engineering, 2023). Natural fiber composites (NFCs) are a class of materials that combine natural fibers with a polymer matrix to enhance mechanical performance and create a composite material with unique properties. Abaca, also known as Manila hemp, is a species of banana, Musa textilis, endemic to the Philippines. The fiber strands average 1 to 3 meters (3.3 to almost 10 feet) in length, depending on petiole size and the processing method used. The lustrous fiber ranges in color from white through brown, red, purple, or black, depending on plant variety and stalk position; the strongest fibers come from the outer sheaths (Britannica Editors, 2026). Abaca fiber is valued for its exceptional strength, flexibility, buoyancy, and resistance to damage in saltwater and UV exposure better than many natural fibers, making it suitable for diverse environments. Pineapple (Ananas comosus) leaf fibers (PALF), as a by-product of one of the largest productive systems in the agro-industrial field, appear as very promising for use in composites and for prospective applications in several fields, such as building and automotive, Before comparing specific materials, it is important to define the core engineering factors that determine whether a material is truly suitable for a drone frame in real operating conditions, rather than just performing well on paper. A suitable frame material is not chosen based on a single property, but on how well it balances multiple performance requirements under flight and manufacturing conditions (2026 Beska LLC USA). Carbon fiber remains the industry standard for UAV frame design because it offers the best combination of stiffness, weight efficiency, and structural performance currently available in scalable manufacturing (sciencedirect.com)

Glass fibers (fiberglass) are super-thin, flexible strands of spun glass. Manufacturers make them by melting sand and minerals at extreme heat, then pulling the liquid glass through tiny holes. It is a strong, lightweight material that does not rot, burn, or conduct electricity. However, despite its advantages, with glass fibers and carbon fibers, Natural fibers are much less expensive and have a lower density. Although the strength of natural fibers is lower, the specific properties are not much different. The rapid advancement of unmanned aerial vehicles (UAVs), commonly known as drones, has transformed numerous industries, including agriculture, environmental monitoring, disaster management, aerial photography, infrastructure inspection, logistics, and military operations. Their ability to perform tasks efficiently, safely, and at a relatively low cost has made them an essential technological tool in both commercial and research applications. As drone technology continues to evolve, there is an increasing demand for lightweight, durable, and high-performance structural materials that can improve flight efficiency, increase payload capacity, reduce energy consumption, and extend flight time. Among the critical structural components of a drone is its frame, which serves as the main support structure that houses and protects essential components such as the motors, battery, flight controller, sensors, and communication systems. Consequently, the selection of an appropriate material for drone frame construction has become a significant consideration in aerospace and mechanical engineering. Traditionally, drone frames are manufactured from aluminum alloys, carbon fiber-reinforced polymers, and fiberglass composites because of their excellent mechanical properties and structural reliability. Carbon fiber, in particular, is widely recognized for its exceptional strength-to-weight ratio and rigidity. However, these materials are often expensive, energy-intensive to manufacture, and difficult to recycle. The increasing concern for environmental sustainability and the depletion of non-renewable resources have encouraged researchers to explore alternative materials that can provide comparable mechanical performance while minimizing environmental impact. As a result, natural fiber-reinforced polymer composites have gained considerable attention as potential substitutes for conventional synthetic composites. Natural fiber composites are composed of biodegradable plant fibers embedded within a polymer matrix. These materials offer several advantages, including low density, good specific strength, renewability, and biodegradability.

STATEMENT OF THE PROBLEM

This study aims to compare the mechanical performance of abaca fiber-reinforced epoxy composites and pineapple fiber-reinforced epoxy composites to determine their potential as lightweight materials for drone frame plates.

1. What are the properties of abaca and pineapple fiber composites in terms of:

    a. weight
    b. tensile strength
    c. flexural strength
    d. impact resistance
    e. structural performance

2. How does the different ratio of abaca and pineapple leaf fibers affect the mechanical and physical properties as a potential application of a composite drone frame plate?

3. How do abaca fiber and pineapple fiber composites perform under different drone operation environments?

4. Which composite material, abaca fiber or pineapple fiber, provides better overall performance for developing a lightweight drone frame under different drone operating environments?

OBJECTIVES OF THE STUDY

This study generally aims to test the potential of Natural fiber composites, between abaca and pineapple leaf fibers, in applications where lightweight, high-strength frames are essential for performance and efficiency.

1. To compare abaca and pineapple leaf fiber quality for making drone frame plates.

2. To determine the effects of varying ratios of abaca and pineapple leaf fibers on the mechanical and physical properties for potential application as drone frame plates.

3. Evaluate the structural and environmental viability of abaca and pineapple leaf fiber composites as lightweight materials for drone frames.

4. To compare the mechanical, physical, and thermal performance of Abaca fiber and Pineapple leaf fiber (PALF) reinforced polymer composites for the development of lightweight drone airframes across varying operating environments.

SIGNIFICANCE OF THE STUDY

This study is significant because it contributes to the development of sustainable and lightweight composite materials for drone frame plate applications. By evaluating the mechanical, physical, and environmental performance of abaca fiber and pineapple leaf fiber (PALF) reinforced polymer composites, the study provides valuable information on the potential of natural fibers as alternatives to conventional synthetic materials. The findings of this study will benefit the following:

Drone Manufacturers and Designers

The results may provide useful information in selecting lightweight, durable, and environmentally friendly composite materials for drone frame plates, which can improve drone performance, fuel or battery efficiency, and payload capacity.

Engineers and Material Scientists

This study may serve as a reference in the development and evaluation of natural fiber-reinforced polymer composites for structural and engineering applications. The findings may also contribute to future innovations in sustainable composite materials.

Farmers and the agricultural sector of abaca and pineapple

The study may help to increase demand for these natural fibers and could create additional economic opportunities for farmers and support the sustainable use of agricultural resources.

Environment

By promoting the use of renewable and biodegradable natural fibers, the study may contribute to reducing dependence on synthetic fiber-reinforced composites.

Future researchers

The study may serve as a valuable reference for future research related to natural fiber composites, composite fabrication techniques, mechanical property evaluation, and sustainable engineering materials. It may also provide baseline data for studies involving other natural fibers or composite formulations.

SCOPE AND LIMITATIONS OF THE STUDY

This study focuses on evaluating the potential of abaca fiber and pineapple leaf fiber (PALF) reinforced polymer composites as lightweight materials for drone frame plates. Specifically, the study compares the mechanical and physical properties of composites fabricated using a selected fiber ratio method adopted by the researchers to determine their suitability for lightweight drone applications. The research includes the fabrication of composite plates using a selected polymer resin reinforced with abaca fiber, pineapple leaf fiber, and their varying combinations. The fabricated specimens will be subjected to laboratory testing to evaluate their weight, tensile strength, flexural strength, impact resistance, and structural performance. In addition, the composites will be exposed outside to assess their environmental durability. The findings of the study will be used to identify the fiber ratio that provides the best balance of lightweight characteristics, mechanical strength, and durability for drone frame plate applications. Furthermore, the results apply only to the materials, fabrication process, specimen dimensions, and testing conditions used in this study and may not be generalized to all natural fiber composites or drone frame designs.

THEORETICAL FRAMEWORK

This study is anchored on four theories (Figure 1). Explain the behavior, design, and selection of fiber-reinforced polymer composites for engineering applications. These theories provide the foundation for understanding how the type of natural fiber, fiber ratio, and environmental conditions influence the mechanical and physical properties of composite materials used in lightweight drone frame plates.

Rule of Mixtures Theory

P. Voigt, further developed for composite materials by Hill (1963)2qaz

Theory explains how varying the proportions of abaca fiber and pineapple leaf fiber (PALF) influences the weight, tensile strength, flexural strength, impact resistance, and structural performance of the fabricated composite. Since the research investigates different fiber ratios, the Rule of Mixtures provides the theoretical basis for understanding how these changes affect the overall performance of drone frame plates.

Bio-Composite Performance Framework

M. Jawaid, S. M. Sapuan, and colleagues (2023). The Bio-Composite Performance Framework explains that the overall performance of natural fiber composites depends on the interaction between fiber properties, matrix characteristics, fiber loading, and manufacturing conditions. This framework is highly relevant because the research investigates different fiber ratios of abaca and pineapple leaf fiber. It explains why changing the reinforcement composition influences the strength, weight, and durability of the resulting composite.

Lightweight Structural Materials Theory

S. M. Sapuan and M. Jawaid (2024)

The Lightweight Structural Materials Framework states that engineering materials for aerospace and unmanned aerial vehicle (UAV) applications should achieve an optimal balance between low weight, high mechanical strength, impact resistance, and environmental durability while maintaining structural reliability.

This framework closely aligns with the objectives of the study because drone frame plates require lightweight materials with excellent mechanical properties. It supports the evaluation of abaca and (PALF) composites to determine which material provides the best performance under different drone operating environments.

Sustainable Natural Fiber-Reinforced Polymer Composite

M. Jawaid, S. M. Sapuan, and co-authors (2021)

This framework explains that natural fiber-reinforced polymer composites should be designed by balancing lightweight characteristics, mechanical strength, environmental sustainability, and manufacturability.

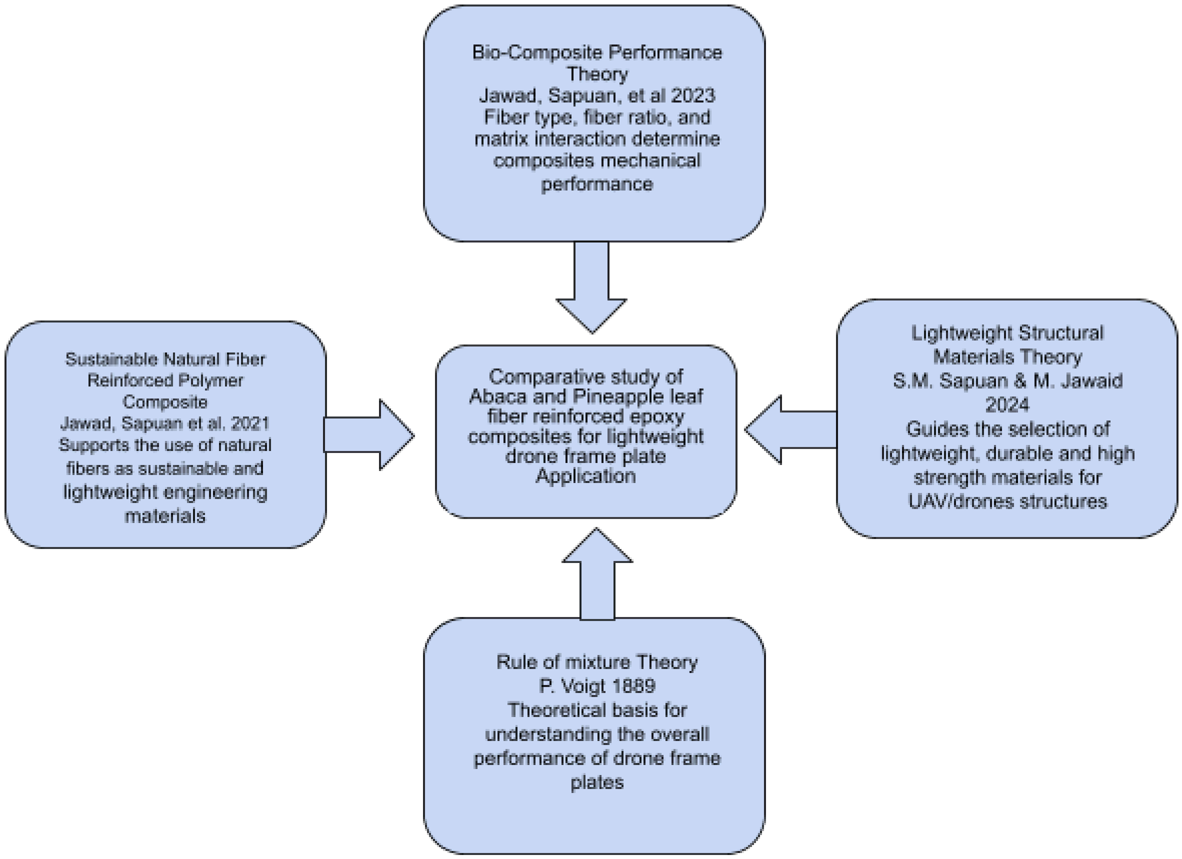

FIGURE 1. THEORETICAL FRAMEWORK

CONCEPTUAL FRAMEWORK

The conceptual framework of this study illustrates the relationship between the independent variables, the research process, and the dependent variables in evaluating natural fiber-reinforced polymer composites for lightweight drone frame plates. The independent variables consist of the fiber type (abaca fiber and pineapple leaf fiber), the fiber ratio (different proportions of abaca and pineapple leaf fiber), and the operating environment (such as ambient conditions, high temperature, and high humidity). These variables are deliberately manipulated to determine their effects on the performance of the composite material. The process begins with the preparation of the natural fibers, followed by the fabrication of the composite plates using a polymer resin. The fabricated composites undergo curing and specimen preparation before being subjected to mechanical and environmental testing. The collected data are then analyzed and compared to determine the influence of the independent variables on the performance of the composite materials. The dependent variables are the measurable properties of the fabricated composites, including weight, tensile strength, flexural strength, impact resistance, structural performance, and environmental durability. These properties serve as the basis for evaluating the suitability of each composite material for drone frame plate applications.

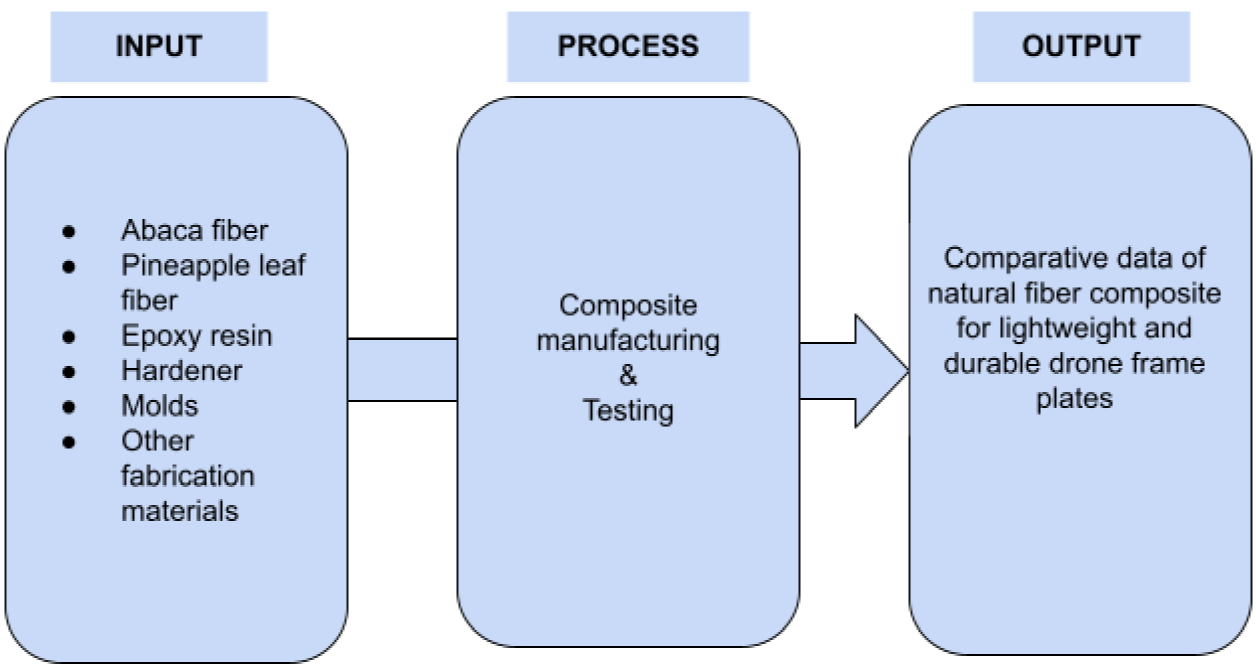

FIGURE 2. CONCEPTUAL FRAMEWORK

Title: ***Mechanical Properties of Natural Fiber-Reinforced Polymer Composites: A Review***

Authors: Faruk, O., Bledzki, A. K., Fink, H. P., & Sain, M.

Year: 2012

Source: ***Progress in Polymer Science, 37(11), 1552–1596.***

Relation to Your Conceptual Framework:

This explains that the independent variables (fiber type, fiber content/ratio, fiber treatment, and matrix material) influence the mechanical properties of natural fiber-reinforced polymer composites through the fabrication and testing process. The dependent variables include tensile strength, flexural strength, impact resistance, stiffness, and density. The study supports a conceptual framework in which the input variables are processed through composite fabrication and mechanical testing to evaluate composite performance.

DEFINITION OF TERMS

Abaca Fiber

Conceptually, abaca fiber is a natural fiber obtained from the leaf sheath of the Musa textilis plant. It is known for its high tensile strength, durability, and resistance to moisture, making it suitable for composite reinforcement. Operationally, it refers to the natural fiber used as a reinforcing material in the fabrication of polymer composite drone frame plates in this study.

Pineapple Leaf Fiber (PALF) (Ananas comosus)

Natural cellulose fiber extracted from the leaves of the pineapple plant. It is recognized for its high strength, low density, and biodegradability. Operationally, it refers to the reinforcing fiber used alone or in combination with abaca fiber in producing composite drone frame plates.

Composite Material

Material formed by combining two or more distinct materials to obtain improved mechanical and physical properties. Operationally, it refers to the polymer composite consisting of a resin matrix reinforced with abaca fiber, pineapple leaf fiber, or a combination of both.

Drone Frame Plate

Conceptually, a drone frame plate is the structural component of an unmanned aerial vehicle (UAV) that supports and connects major parts such as the motors, battery, and flight controller. Operationally, it refers to the fabricated composite plate evaluated as a potential lightweight material for drone frame construction.

Fiber Ratio

The proportion of two or more reinforcing fibers within a composite material. Operationally, it refers to the different proportions of abaca fiber and pineapple leaf fiber used in fabricating the composite specimens.

Structural Performance

Refers to the ability of a material or structure to withstand applied loads while maintaining its integrity and functionality. Operationally, it refers to the overall mechanical behavior of the composite specimens based on the results of weight, tensile, flexural, and impact tests.

Environmental Durability

Ability of a material to maintain its performance when exposed to environmental conditions such as heat and humidity. Operationally, it refers to the performance of the fabricated composites after exposure to the selected drone operating environments.

Lightweight Material

Material with relatively low density that provides sufficient strength for structural applications. Operationally, it refers to the composite material identified as having the

best combination of low weight and high mechanical performance for drone frame plates.

Mechanical Properties

Essential for ensuring the performance and safety standards of materials in various engineering applications.

Weight efficiency

A lighter structure improves flight endurance and reduces power consumption, especially in long-range or battery-sensitive UAV systems.

Rigidity

Structural stiffness determines how well the frame maintains geometric stability under aerodynamic loads and high-speed maneuvers.

Vibration behavior

Good vibration damping helps protect onboard sensors and improves control accuracy, particularly in imaging and inspection drones.

Machinability

From a manufacturing perspective, the material must be suitable for processes such as CNC machining, cutting, drilling, and surface finishing with consistent tolerance.

Corrosion resistance

Outdoor UAVs often operate in humid, coastal, or chemically exposed environments, making material stability over time a critical factor.

Natural fibers

Simple fibers that are made or not synthetic. Can be found in plants and animals; the use of natural fiber from both resources is renewable and non-renewable.

Chapter II

METHODOLOGY

This study evaluates the mechanical performance of Abaca and pineapple leaf fiber (PALF) epoxy composites for drone frames. Methodology includes fiber alkali treatment, composites fabrication via hand lay-up, and mechanical testing. This study aims to identify the optimum natural fiber for lightweight, vibration-damping aerospace components.

RESEARCH DESIGN

The researcher utilizes a comparative experimental research design to evaluate, fabricate, test, and compare the performance of Abaca and Pineapple Leaf Fiber (PALF)-reinforced epoxy composites in different fiber ratios. The goal is to determine the most mechanically robust and lightweight biocomposite formulation to replace traditional synthetic materials in drone frame plates.

RESEARCH METHOD

This study employed an experimental comparative design to evaluate how abaca and pineapple leaf fibres perform as structural reinforcements in epoxy-based drone frames. To determine the optimal material for lightweight, structural UAV Components

DATA GATHERING PROCEDURES

The researchers will fabricate composite drone frame plate specimens using abaca fiber and pineapple leaf fiber (PALF) with varying fiber-reinforcement ratios. To ensure the validity of the results, all specimens will be prepared under identical fabrication and curing conditions. Each specimen will then undergo standardized laboratory testing to determine its weight, tensile strength, flexural strength, impact resistance, and structural performance. Furthermore, the fabricated composites will be subjected to different simulated drone operating environments, including heat, humidity, and vibration, to evaluate their environmental performance and durability. The data obtained from each test will be systematically recorded, organized, and prepared for statistical analysis.

STATISTICAL TREATMENT OF DATA

The data collected from the experimental tests were analyzed using descriptive and inferential statistical methods to compare the performance of varying abaca and pineapple fiber ratios and drone operating environments on the physical and mechanical properties of the composite materials, specifically in terms of weight, tensile strength, flexural strength, impact resistance, and structural performance, for lightweight drone frame plate applications. To compute the statistical treatment of data, researchers used mean, standard deviation, and two-Way Analysis of Variance (ANOVA).

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